Investigating the Relationship Between Math Enjoyment and EQAO Results

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Abstract

This study explores the relationship between the EQAO achievement and the math enjoyment among 246 elementary school and secondary schools within the Toronto District School Board in the year of 2024. Data were sourced from the publicly available reports on the Education Quality and Accountability Office (EQAO) website. Through the scatter plot analysis and correlation coefficient calculations of the elementary school and the secondary school, The findings both reveal a moderate positive correlation between students' math enjoyment and their EQAO performance, suggesting that students who report greater engagement in math tend to achieve higher standardized assessment scores. The results highlight the importance of fostering positive attitudes towards mathematics in educational practices. Encouraging enjoyment of math may be a key strategy for enhancing overall student performance in standardized testing and promoting long-term success in mathematical learning.

Introduction

Mathematics scores have always been an important indicator for people to measure a student's academic performance and future work success. In Ontario, the relevant Ministry of Education uses the EQAO mathematics test, which is divided into two grades, Grade 6 and Grade 9, to assess the learning outcomes of students in these two grades. Although standardized test scores are often used to measure the effectiveness of teaching strategies and curriculum design, people are increasingly aware that the role of mathematics fun (students' positive attitude and enthusiasm for mathematics) is gaining more and more attention. The fun of mathematics is usually defined as students' participation in mathematics, their study of questions, their interest and sense of reward. A positive attitude towards mathematics is believed to promote deeper learning and sustained investment. Previous studies have shown that students who like mathematics are more likely to actively participate in classroom activities, spend time practicing skills, and achieve higher academic achievements. In other words, it is a virtuous circle and positive correlation. Therefore, this study focuses on the relationship between math interest and EQAO scores in 246 elementary and secondary schools under the Toronto District School Board (TDSB). Based on public data from the EQAO website, by analyzing the percentage of students interested in math and the percentage of test grades above the provincial standards, it aims to gain a clearer understanding of how students' math attitudes affect their standardized test scores, thereby highlighting the importance of cultivating a positive math attitude in educational practice.

Methods

This study aims to analyze the relationship between the proportion of students who scored above the provincial standard in the EQAO math test in 2024 and the proportion of students who are interested in math. In order to reduce the bias of the survey data, this study only investigated the data of grades 6 and 9. Because grade 3 may not be very mature in self-cognition, it may cause some bias and extreme values in the data of their self-report to the enjoyment of math. Considering the authenticity, reliability and randomness of the data survey, this study selected a total of 246 elementary and secondary schools under TDSB that held EQAO exams through the EQAO official website, excluding some schools that use special exam questions and do not hold EQAO exams. This ensures high coverage of the target population.

The two important variables in this survey are Math Enjoyment and EQAO Score. This variable represents the degree to which students enjoy math. This variable is based on the student attitude survey conducted by EQAO and is represented by the percentage of students in each school who agree that he/she "likes" or "enjoys" math. It reflects the students' subjective feelings and positive attitudes towards learning math. EQAO Score: This variable represents the percentage of students in each school who scored above the provincial standard on the math standardized test.

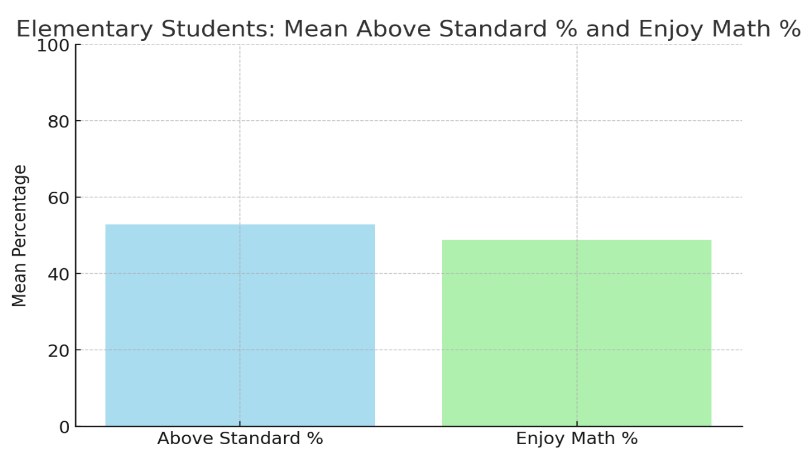
First, for both elementary schools and secondary schools, this study used Google Sheets to record the percentage of sixth- and ninth-grade math enjoyment and scores above the provincial standard for each school to be surveyed. Next, by using the formula for calculating the average, calculate the average percentage of elementary and secondary school students whose EQAO test grades are above the provincial standard and the average percentage of students who recognize that they "enjoy math". After that, use the formula for calculating standard deviation to calculate the standard deviation of elementary and secondary school students whose EQAO test grades are above the provincial standard and the standard deviation of students who recognize that they "enjoy math". These data facilitate the calculation of correlation coefficients and the discussion and analysis of potential trends and causes later in this study.

Next, use Google Sheets to draw the corresponding scatter plot, and then connect them into a roughly straight line. Next, use the formula for calculating the correlation coefficient to calculate the corresponding correlation value. Finally, based on the corresponding values, the correlation between the percentage of EQAO test grades above the provincial standard and the percentage of students who "enjoy mathematics" in this study is obtained.

Results

After the graphing and the calculations, the results are shown below:

For Elementary Schools:



Mean of the student above standard:

$$\bar{x} = \frac{1}{n}\Sigma x_i = 52.94\%$$

Mean of the student enjoy math:

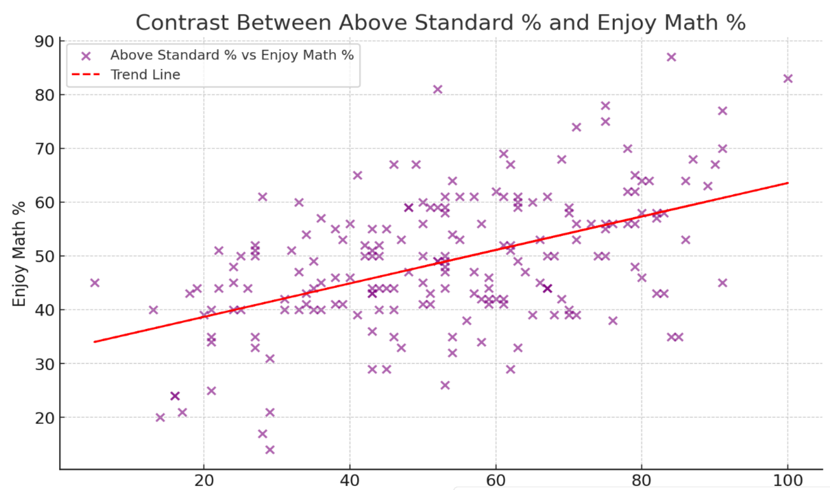
$$\bar{x} = \frac{1}{n}\Sigma x_i = 48.9\%$$

Standard deviation of student above standard:

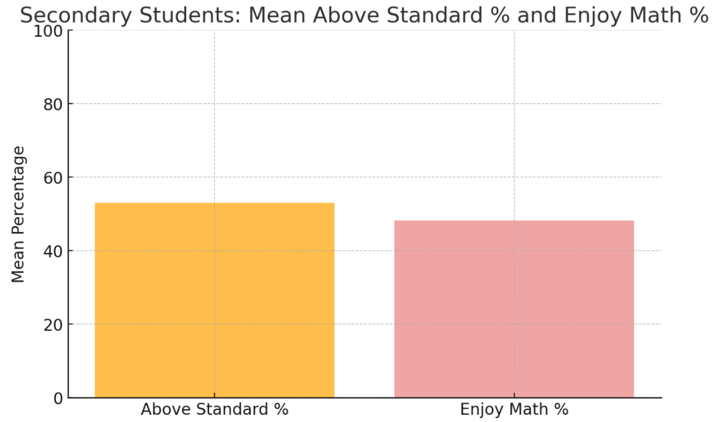
$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{77675.26}{195-1}} = 20.01$$

Standard deviation of student who enjoy math:

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{30143.16}{195-1}} = 12.47$$



Correlation coefficient: $$r = \frac{\Sigma(x-\bar{x})\ (y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2\ \Sigma(y-\bar{y})^2}} = \frac{24160.08}{48375.49} = 0.4994$$



For Secondary Schools:

Mean of the student above standard:

$$\bar{x} = \frac{1}{n}\Sigma x_i = 53.02\%$$

Mean of the student enjoy math:

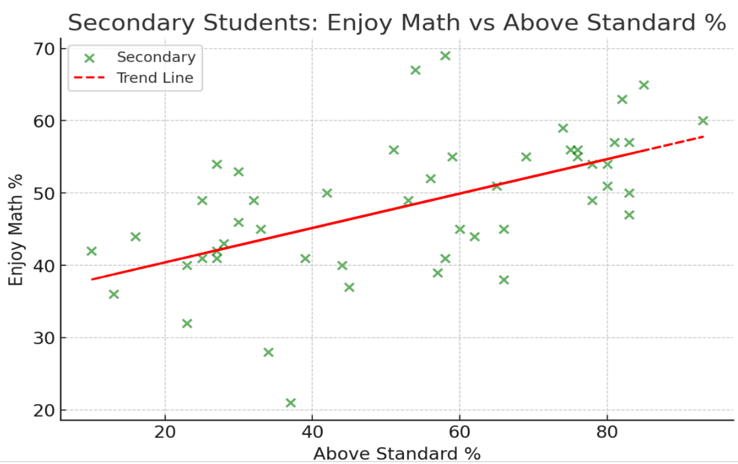
$$\bar{x} = \frac{1}{n}\Sigma x_i = 48.26\%$$

Standard deviation of student above standard:

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{26440.98}{51-1}} = 22.99$$

Standard deviation of student who enjoy math:

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{4573.62}{51-1}} = 9.56$$



Correlation coefficient: $r = \frac{\Sigma(x-\bar{x})\ (y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2\ \Sigma(y-\bar{y})^2}} = \frac{6292.74}{10996.86} = 0.5722$

Data calculation summary:

Elementary Schools:

	Mean percentage (1-variable analysis)	Standard deviation (1-variable analysis)	Correlation coefficient (2-variable analysis)
Percentage above provincial standard	52.94%	20.01	0.4994
Math enjoyment	48.9%	12.47	0.4994

Secondary Schools:

	Mean percentage (1-variable analysis)	Standard deviation (1-variable analysis)	Correlation coefficient (2-variable analysis)
Percentage above provincial standard	53.02%	22.99	0.5722
Math enjoyment	48.26%	9.56	0.5722

Discussion

Elementary schools discussion:

Results from this analysis indicate a moderate positive correlation between elementary school students' enjoyment of mathematics and their performance on the EQAO assessment. This finding suggests that, in general, schools with a higher percentage of students reporting enjoyment of mathematics have a higher percentage of EQAO test scores exceeding the provincial standard for mathematics. However, the analysis also revealed some extreme values in the data set. For example, at Essex Junior and Senior Public School, the school reported significantly lower levels of enjoyment of mathematics at 14%, but had a similar percentage of students scoring above the provincial standard as the other schools. Conversely, at Cedar Drive Junior Public School, the school reported high levels of enjoyment of mathematics at 61%, but had a relatively low percentage of students scoring above the provincial standard at 28%. These extremes could be influenced by a variety of factors outside the scope of this analysis, including differences in the quality of instruction, school resources, or self-reported indicators of enjoyment of mathematics. The math enjoyment data relies on students' self-reported attitudes, which can be affected by social desirability bias, different interpretations of survey questions, and respondents' transient emotional states. Second, it does not account for differences in socioeconomic status, language diversity, or other contextual factors that could affect math enjoyment and test performance. Moreover, since this study only analyzed the year of 2024, it does not account for changes over time or trends across multiple years.

Secondary schools discussion:

Based on data from secondary schools, there was a moderate positive correlation between students' reported enjoyment of math and their performance on the EQAO assessment. Despite this general trend, as with elementary schools, there were some extreme values in secondary schools. For example, at Oakwood Collegiate Institute, the level of enjoyment of math was significantly lower at 21%, but the proportion of students achieving math scores above the provincial standard was similar to other schools. Conversely, at Parkdale Collegiate Institute, the level of enjoyment of math was high at 67%, but the proportion of students achieving the provincial standard was relatively low for schools of this interest level, at 58%. However, this study only had data from 2024 and did not capture longitudinal trends or changes in attitudes and performance over time. The moderate correlations found do not establish causality. Potential biases may also have influenced the results. The math enjoyment data were self-reported by students, which introduces the possibility of social desirability bias or student misunderstanding of the questions.

Comparison and summary between primary and secondary schools:

In comparison between elementary and secondary schools, the average percentage of EQAO math test grades above the provincial standard and the average percentage of students' "math enjoyment" are almost the same. However, the standard deviation of percentage above provincial standard for secondary school students is slightly higher than that for elementary school students. This may be because secondary schools usually offer more specialized math pathways, which may lead to greater differences in grades between students. In addition, the academic backgrounds of secondary school students may be more diverse, affected by factors such as different elementary school math grades and extracurricular support. In addition, the standard deviation of percentage of math enjoyment for secondary school students is slightly lower than that for elementary school students. This may be due to the increasing academic pressure and more standardized expectations in secondary education. Secondary school students may also have a more realistic or stable view of mathematics based on years of accumulated experience, resulting in less difference in their enjoyment responses. Moreover, it can be noted that the overall percentage of enjoyment of math for secondary school students is lower than that for elementary school students. The largest one does not exceed 70%, but the largest one in elementary school is 87%. Possible reasons are increased academic pressure and complexity: As students enter secondary school, math courses become more challenging and abstract. Concepts such as algebra, functions, and geometry require higher levels of cognitive processing and problem-solving skills, which may reduce the interest of some students who have difficulty mastering these complex topics. However, the common point between both secondary and elementary schools is that there is a moderate positive correlation between enjoyment in math and achievement above the provincial standard. The possible reason should be that enjoyment and achievement might feed off each other in a virtuous cycle. Since performance builds confidence: When students consistently meet or exceed the standard, they feel more confident in math and thus report enjoying it more. Given the moderate positive correlation between enjoyment and achievement in mathematics in both primary and secondary schools, future efforts should focus on developing students' positive attitudes toward mathematics, and schools and educators can provide relevant and supportive learning environments to foster students' confidence and interest in mathematics. Recognizing that developing positive attitudes toward mathematics is a key driver of academic success, schools can implement classroom activities to enhance students' engagement and sense of achievement in mathematics.

References

Education Quality and Accountability Office. (2024). Assessments and results. EQAO. Retrieved July 9, 2025, from https://www.eqao.com/

Toronto District School Board. (2023). Parent and student census. TDSB Research. Retrieved July 9, 2025, from https://www.tdsb.on.ca/research/Research/Parent-and-Student-Census

Appendix

Raw data table link:

https://docs.google.com/spreadsheets/d/1haCET_r72iQeqcpCoxG8wQjPE3heFqNubyTOpXzZiTg/edit?gid=0#gid=0

Calculation data summary:

Elementary Schools:

	Mean percentage (1-variable analysis)	Standard deviation (1-variable analysis)	Correlation coefficient (2-variable analysis)
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Secondary Schools:

	Mean percentage (1-variable analysis)	Standard deviation (1-variable analysis)	Correlation coefficient (2-variable analysis)
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Math enjoyment	48.26%	9.56	0.5722